

4.1 Write a program to implement Depth First Search using linked representation of graph.

```
import java.util.*;
```

```
class GraphDFS {
```

```
    int V;
```

```
    LinkedList<Integer> adj[];
```

```
    GraphDFS(int v) {
```

```
        V = v;
```

```
        adj = new LinkedList[v];
```

```
        for (int i = 0; i < v; i++)
```

```
            adj[i] = new LinkedList<Integer>();
```

```
    }
```

```
    void addEdge(int v, int w) {
```

```
        adj[v].add(w);
```

```
    }
```

```
    void DFSUtil(int v, boolean visited[]) {
```

```
        visited[v] = true;
```

```
        System.out.print(v + " ");
```

```
        Iterator<Integer> i = adj[v].listIterator();
```

```
        while (i.hasNext()) {
```

```
            int n = i.next();
```

```
            if (!visited[n])
```

```
        DFSUtil(n, visited);
    }
}
```

```
void DFS(int v) {
    boolean visited[] = new boolean[V];
    DFSUtil(v, visited);
}
```

```
public static void main(String args[]) {
    GraphDFS g = new GraphDFS(5);

    g.addEdge(0, 1);
    g.addEdge(0, 2);
    g.addEdge(1, 3);
    g.addEdge(2, 4);

    System.out.println("Depth First Search:");
    g.DFS(0);
}
```

```
1 warning
Depth First Search:
0 1 3 2 4
PS C:\Users\nbehe\Music\MCA\SEM2\dsa_lab\lab4>
Indexing completed. Java: Ready
```

4.2 Write a program to implement Breadth First Search using linked representation of graph.

```
import java.util.*;
```

```
class GraphBFS {
```

```
int V;
```

```
LinkedList<Integer> adj[];
```

```
GraphBFS(int v) {
```

```
    V = v;
```

```
    adj = new LinkedList[v];
```

```
    for (int i = 0; i < v; i++)
```

```
        adj[i] = new LinkedList<Integer>();
```

```
}
```

```
void addEdge(int v, int w) {
```

```
    adj[v].add(w);
```

```
}
```

```
void BFS(int s) {
```

```
    boolean visited[] = new boolean[V];
```

```
    LinkedList<Integer> queue = new LinkedList<Integer>();
```

```
    visited[s] = true;
```

```
    queue.add(s);
```

```
    while (queue.size() != 0) {
```

```
        s = queue.poll();
```

```
        System.out.print(s + " ");
```

```
        Iterator<Integer> i = adj[s].listIterator();

        while (i.hasNext()) {
            int n = i.next();
            if (!visited[n]) {
                visited[n] = true;
                queue.add(n);
            }
        }
    }
}
```

```
public static void main(String args[]) {
```

```
    GraphBFS g = new GraphBFS(5);
```

```
    g.addEdge(0, 1);
```

```
    g.addEdge(0, 2);
```

```
    g.addEdge(1, 3);
```

```
    g.addEdge(2, 4);
```

```
    System.out.println("Breadth First Search:");
```

```
    g.BFS(0);
```

```
}
```

```
1 warning
Breadth First Search:
0 1 2 3 4
PS C:\Users\nbehe\Music\MCA\SEM2\dsa_lab\lab4>
6 Indexing completed. Java: Ready
```